Mathematical Methods for Finance

Module 2 – Lebesgue Integration and Convergence Theorems

Lesson 7: Integrals, limits and theorem selection

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 7

By the end of this lesson, students should be able to:

1. compute integrals of nonnegative simple functions;
2. explain the construction idea of the Lebesgue integral;
3. distinguish integrable functions from merely measurable functions;
4. use MCT, Fatou and DCT in basic examples;
5. recognize when Fubini or Tonelli can be applied.

Exam relevance

This lesson provides the language needed to read the next modules: measurable sets, measurable functions and measures are the base layer of integration and probability.

How today is organised

14:30–15:20

Concepts and intuition
From area to Lebesgue integration.

15:35–16:25

Theory and examples
Integrability and convergence theorems.

16:40–17:30

Exercises
Choose the correct theorem and compute.

Intuition: Learning rule

For each topic: identify the object, write the definition, test it on a small example, then solve one exam-style exercise.

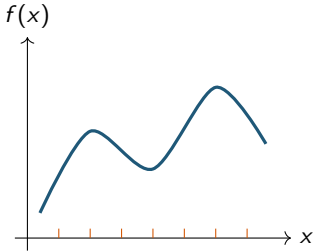
Module 2

Block 1: from simple functions to integrals

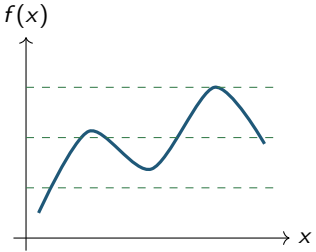
Why a new integral?

Intuition: Riemann versus Lebesgue

Riemann integration partitions the domain. Lebesgue integration groups the values of the function.



Riemann: split the input axis



Lebesgue: group values

Integral of a nonnegative simple function

Definition: Simple integral

Let

$$s = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}, \qquad a_k \geq 0,$$

where the A_k are measurable and disjoint. Define

$$\int_X s \, d\mu = \sum_{k=1}^{m} a_k \mu(A_k).$$

Intuition: Meaning

Height times size, summed over the pieces.

Worked example: finite simple integral

Worked example: Three-point space

Let $X = \{1, 2, 3\}$ with counting measure and

$$s(1) = 2, \qquad s(2) = 5, \qquad s(3) = 1.$$

Then

$$\int_X s \, d\mu = 2 + 5 + 1 = 8.$$

Exercise: Weighted version **

If the weights are $\mu(\{1\}) = 1$, $\mu(\{2\}) = 3$, $\mu(\{3\}) = 2$, compute $\int s \, d\mu$.

Solution: weighted simple integral

Proof idea: Computation

$$\int_X s \, d\mu = s(1)\mu(\{1\}) + s(2)\mu(\{2\}) + s(3)\mu(\{3\}).$$

Thus

$$\int_X s \, d\mu = 2 \cdot 1 + 5 \cdot 3 + 1 \cdot 2 = 19.$$

Nonnegative measurable functions

Definition: Lebesgue integral, construction idea

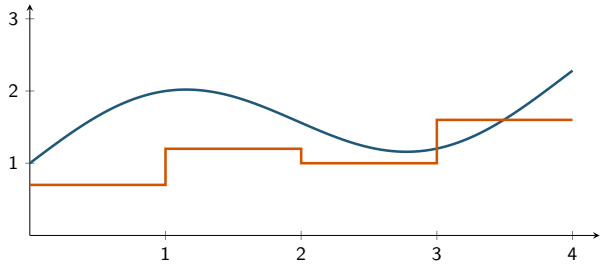
If $f \geq 0$ is measurable, define

$$\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu : 0 \leq s \leq f, \ s \text{ simple} \right\}.$$

Intuition: Meaning

Approximate f from below by step functions, integrate the step functions, then take the supremum.

Picture: approximation from below



Intuition: Message

Simple functions are not a toy example: they are the foundation of the integral.

General integrable functions

Definition: Positive and negative parts

For a measurable function f, define

$$f^+ = \max(f, 0), \qquad f^- = \max(-f, 0).$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Definition: Integrable function

A measurable function f is integrable if

$$\int_X |f| \, d\mu < \infty.$$

Then

$$\int_X f \, d\mu = \int_X f^+ \, d\mu - \int_X f^- \, d\mu.$$

Absolute value and integrability

Intuition: Why absolute value returns

To integrate signed functions safely, positive and negative parts cannot both be infinite. The condition $\int |f| < \infty$ prevents this.

Example: On $[0, 1]$

The function $f(x) = x$ is integrable and

$$\int_0^1 |x| \, dx = \frac{1}{2}.$$

Counterexample: Not integrable

The function $g(x) = 1/x$ on $(0, 1)$ is not integrable with respect to Lebesgue measure because

$$\int_0^1 \frac{1}{x} \, dx = \infty.$$

Basic integral properties

Theorem: Linearity and monotonicity

If f, g are integrable and $a, b \in \mathbb{R}$, then

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu.$$

If $f \leq g$ almost everywhere, then

$$\int f\, d\mu \leq \int g\, d\mu.$$

Exercise: Quick use **

If $\int f\, d\mu = 2$ and $\int g\, d\mu = -1$, compute $\int (3f - 2g)\, d\mu$.

Solution: linearity

Proof idea: Computation

$$\int (3f - 2g)\, d\mu = 3 \int f\, d\mu - 2 \int g\, d\mu.$$

Therefore

$$3 \cdot 2 - 2 \cdot (-1) = 6 + 2 = 8.$$

Equivalence classes

Definition: Equality almost everywhere

Two measurable functions f and g are equal almost everywhere if

$$\mu(\{x : f(x) \neq g(x)\}) = 0.$$

Theorem: Integral ignores null changes

If $f = g$ almost everywhere and f is integrable, then

$$\int f \, d\mu = \int g \, d\mu.$$

Future use

This is why L^p spaces are spaces of equivalence classes, not individual pointwise functions.

Module 2

Block 2: convergence theorems

Why convergence theorems matter

Intuition: Main question

When can we exchange limit and integral?

$$\lim_{n\to\infty} \int f_n \, d\mu \quad ? \quad \int \lim_{n\to\infty} f_n \, d\mu.$$

Counterexample: Warning

Pointwise convergence alone is not enough.

Example: Mental picture

Mass can move, concentrate or disappear in the limit. Theorems control this behavior.

Monotone Convergence Theorem

Theorem: MCT

Let $0 \leq f_1 \leq f_2 \leq \cdots$ be measurable functions and suppose $f_n \uparrow f$ pointwise. Then

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

Equivalently,

$$\lim_{n\to\infty} \int f_n \, d\mu = \int \lim_{n\to\infty} f_n \, d\mu.$$

When to think of MCT

Nonnegative increasing sequence of functions.

MCT example

Worked example: Increasing indicators

On $[0,1]$, let

$$f_n(x) = \mathbf{1}_{[0,1-1/n]}(x).$$

Then $f_n \uparrow \mathbf{1}_{[0,1)}$ pointwise and

$$\int_0^1 f_n(x)\,dx = 1 - \frac{1}{n} \to 1.$$

Also

$$\int_0^1 \mathbf{1}_{[0,1)}(x)\,dx = 1.$$

Fatou's lemma

Theorem: Fatou

If $f_n \geq 0$ are measurable, then

$$\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

Intuition: Meaning

Without monotonicity or domination, we may still get a one-sided inequality.

When to think of Fatou

Nonnegative functions, weak information, need an inequality.

Fatou example

Example: Moving spikes

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $(0,1)$. Then $f_n(x) \to 0$ for every fixed $x \in (0,1)$, but

$$\int_0^1 f_n(x)\,dx = 1.$$

Fatou gives

$$0 = \int 0\,dx \leq \liminf_n 1 = 1.$$

Counterexample: Lesson

Pointwise convergence does not force convergence of integrals.

Dominated Convergence Theorem

Theorem: DCT

Let $f_n \to f$ pointwise almost everywhere. Suppose there exists an integrable function g such that

$$|f_n| \leq g \quad \text{almost everywhere for all } n.$$

Then f is integrable and

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

When to think of DCT

Pointwise convergence plus one integrable bound dominating all functions.

DCT example

Worked example: A bounded oscillating sequence

On $[0,1]$, let

$$f_n(x) = x^n.$$

Then $f_n(x) \to 0$ for $x \in [0,1)$ and $f_n(1) = 1$. Thus $f_n \to 0$ almost everywhere. Also

$$0 \le f_n(x) \le 1,$$

and $g(x) = 1$ is integrable on $[0,1]$. By DCT,

$$\int_0^1 x^n dx \to 0.$$

Indeed $\int_0^1 x^n dx = 1/(n+1)$.

Decision table: which theorem?

Situation	Tool
$f_n \geq 0$ and $f_n \uparrow f$	Monotone Convergence Theorem
$f_n \geq 0$ and no monotonicity	Fatou's lemma
$f_n \to f$ and $\lvert f_n \rvert \leq g \in L^1$	Dominated Convergence Theorem
Pointwise convergence only	Not enough

Exercise: Exam skill ***

For each problem, first write which theorem you want to use and which hypothesis you must verify.

Exercise: choose the theorem

Exercise: Theorem selection ***

On $[0,1]$, consider

$$f_n(x) = \frac{x}{1+nx}.$$

1. Find the pointwise limit.
2. Find a simple dominating function.
3. Conclude the limit of the integrals.

Solution: choose the theorem

Proof idea: DCT

For $x = 0$, $f_n(0) = 0$. For $x > 0$,

$$\frac{x}{1 + nx} \to 0.$$

Hence $f_n \to 0$ pointwise. Also

$$0 \leq f_n(x) = \frac{x}{1 + nx} \leq x \leq 1.$$

The function $g(x) = 1$ is integrable on $[0, 1]$. By DCT,

$$\int_0^1 \frac{x}{1 + nx}\, dx \to 0.$$

Exercise: MCT or DCT?

Exercise: Increasing sequence ***

On $[0,1]$, let

$$f_n(x) = \min(nx, 1).$$

1. Show that $f_n \uparrow 1$ for $x > 0$.
2. Use MCT to compute the limit of $\int_0^1 f_n(x)\,dx$.

Solution: MCT

Proof idea: Increasing sequence

For fixed $x > 0$, $nx \to \infty$, so $\min(nx, 1) \to 1$. At $x = 0$, $f_n(0) = 0$. Thus $f_n \to 1$ almost everywhere. Moreover f_n is increasing in n and $f_n \geq 0$. By MCT,

$$\lim_n \int_0^1 f_n \, dx = \int_0^1 1 \, dx = 1.$$

Module 2

Block 3: product measures and Fubini-Tonelli

Product spaces

Definition: Product space

Given measurable spaces $(X, \mathcal{F})$ and $(Y, \mathcal{G})$, the product sigma-algebra $\mathcal{F} \otimes \mathcal{G}$ is generated by measurable rectangles

$$A \times B, \qquad A \in \mathcal{F}, \quad B \in \mathcal{G}.$$

Intuition: Meaning

To integrate functions of two variables, we need a measurable structure on the product space.

Product measure

Definition: Product measure

If μ is a measure on $(X, \mathcal{F})$ and ν is a measure on $(Y, \mathcal{G})$, the product measure $\mu \otimes \nu$ satisfies

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$$

for measurable rectangles.

Example: Lebesgue measure on the plane

The product of length with length gives area:

$$\lambda^2([a, b] \times [c, d]) = (b - a)(d - c).$$

Tonelli theorem

Theorem: Tonelli

If $f : X \times Y \to [0, +\infty]$ is measurable, then

$$\int_{X\times Y} f \, d(\mu \otimes \nu) = \int_X \left(\int_Y f(x,y) \, d\nu(y) \right) d\mu(x)$$

$$= \int_Y \left(\int_X f(x,y) \, d\mu(x) \right) d\nu(y).$$

When to use Tonelli

Nonnegative functions. Integrability is not required beforehand.

Fubini theorem

Theorem: Fubini

If f is integrable on $X \times Y$, i.e.

$$\int_{X\times Y} |f|\, d(\mu \otimes \nu) < \infty,$$

then the iterated integrals exist and

$$\int_{X\times Y} f\, d(\mu \otimes \nu) = \int_X \int_Y f(x,y)\, d\nu(y) d\mu(x).$$

When to use Fubini

Signed functions, but you must check absolute integrability.

Worked example: rectangle integral

Worked example: Two variables

Compute

$$\int_0^1 \int_0^2 (x+y)\,dy\,dx.$$

First integrate in y:

$$\int_0^2 (x+y)\,dy = 2x+2.$$

Then integrate in x:

$$\int_0^1 (2x+2)\,dx = 3.$$

Exercise: reverse the order

Exercise: Fubini computation **

Compute the same integral by reversing the order:

$$\int_0^2 \int_0^1 (x + y)\, dx\, dy.$$

Solution: reverse the order

Proof idea: Computation

First integrate in x:

$$\int_0^1 (x+y)\,dx = \frac{1}{2} + y.$$

Then integrate in y:

$$\int_0^2 \left(\frac{1}{2} + y\right) dy = 1 + 2 = 3.$$

The result agrees with the previous order.

Common mistakes

Counterexample: Mistake 1

Using Fubini without checking integrability when the function can be negative.

Counterexample: Mistake 2

Confusing Tonelli and Fubini: Tonelli is for nonnegative functions; Fubini is for integrable signed functions.

Counterexample: Mistake 3

Forgetting that a function must first be measurable.

Lesson 7 integrated exercise

Exercise: Full mini-problem ****

Let

$$f_n(x) = n\mathbf{1}_{(0,1/n)}(x), \qquad x \in (0,1).$$

1. Find the pointwise limit.
2. Compute $\int_0^1 f_n(x)\,dx$.
3. Decide whether DCT applies.
4. Explain what this example teaches.

Solution: integrated exercise

Proof idea: Detailed steps

For every fixed $x > 0$, eventually $x > 1/n$, so $f_n(x) = 0$. Hence $f_n \to 0$ pointwise. But

$$\int_0^1 f_n(x)\,dx = n \cdot \frac{1}{n} = 1.$$

DCT does not apply: there is no integrable function g dominating all spikes. The example shows that pointwise convergence alone does not control integrals.

End of Lesson 7: what to remember

Checklist

1. Simple functions are the building blocks of the integral.
2. Integrability means $\int |f| < \infty$.
3. Null changes do not change the integral.
4. MCT: nonnegative increasing.
5. Fatou: nonnegative, inequality.
6. DCT: pointwise convergence plus integrable domination.
7. Tonelli: nonnegative product integrals.
8. Fubini: integrable signed product integrals.